Module4 Project

TaskM4.T1D: Project

Overview of the task

To fulfill the requirements of this task, you will need to demonstrate your skills to implement a theme++ - something extra in one or more of the themes or modules we discussed in this unit.

Submission Details

Please make sure to provide the following:

- Document of your project scope, design, implementation and evaluation
- Your project code, and
- High-quality 10+ minute video of your solution running.

Instructions

1. Start your project early, share some ideas with the unit team to discuss earlier

2. Plan your project ahead, break down in spikes or sprints as needed

3. Make sure you always have a working prototype to submit if you run out of time

4. Submit your task as detailed on the submission details section above to OnTrack.

Assurance of Learning. Report and Demonstration Requirements

- Teaching staff may ask extra questions in chat and/or in a short interview to confirm understanding and authorship of the project scope, design, build and evaluation.
- The student must run at least one stress test for the project workload that either: (a) does not compile, or (b) is **DNF** (did not finish), or (c) runs for too long (state how long was waited, for example “waited 10 minutes”); the student must state the machine specifications used (CPU model, number of cores/threads, RAM, OS) and, if known, the reason for **DNF** (for example, compile error, out of memory, timeout, or user interrupted due to long wait).
- The student must show an end-to-end working project and must provide exact compile and run commands, plus a screenshot that shows the project running.
- The student must provide evaluation evidence using at least three increasing workload sizes relevant to the project and must explain whether speedup was achieved and why. The student must choose sizes based on their own computer specifications so that runtimes are not too close to compare and the results support clear conclusions.
- If one of the technology stacks from the previous unit materials is used (`threading` libraries, `OpenMP`, `MPI`, `OpenCL`/`Metal`, `Docker` or something from `lecture demos`,), the student should be able to clearly articulate, what new approach is introduced in the Project, that wasn't covered by the unit materials previously and what research was necessary for the student to implemented this `D`/`HD` Project.
- If a video is required, the live demonstration must match the video’s features and code and the student must answer questions beyond what is shown in the video.

Quality Point demonstration tiers (show only the tier being claimed)

Level 1 (working prototype)

- Show a working prototype that matches the stated scope and can be run using only the provided commands and files.

Level 2 (evaluation-led work, lower HD)

- In addition to Level 1, show a clear test method (repeat runs, state the system used and test multiple workload sizes) and give a short academic analysis of results.

Level 3 (advanced work and defence, high HD)

- In addition to Level 2, show clear added work beyond a basic pattern (for example, hybrid parallel work, load balance methods, or stronger test method) and justify key choices under questions. If the Project is an improvement of some other student's projects from other units, the student must clearly explain what was added or improved over the previous version of the project and what concurrent or distributed programming paradigms have been utilised.